Module 1-CSS3

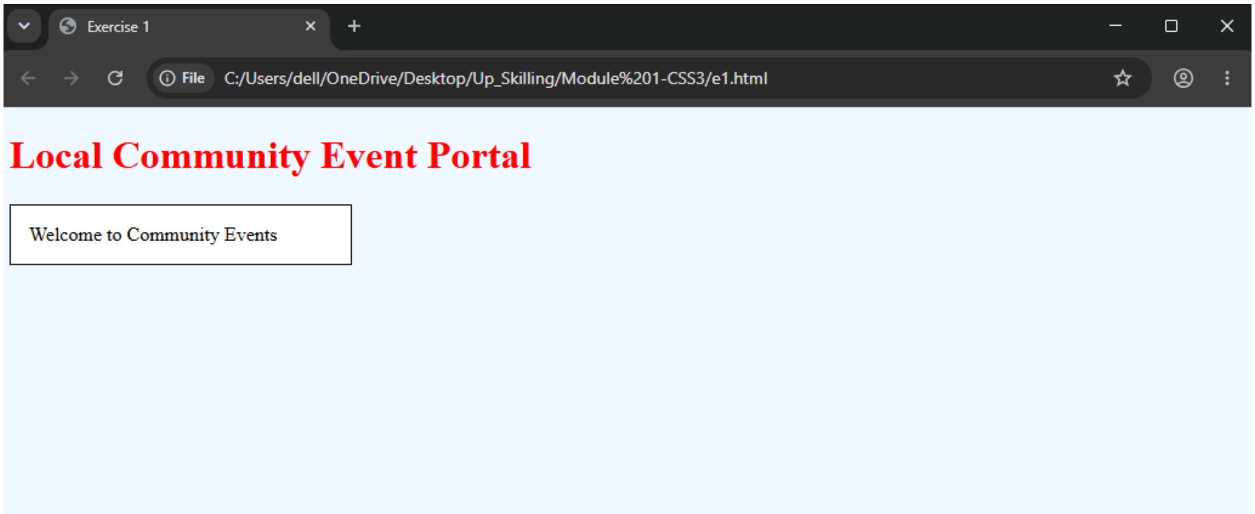
Exercise 1

Problem Statement:
Why CSS? Inline vs. Internal vs. External
Scenario: The designer wants you to experiment with different ways to apply styles.
Objective: Understand various CSS inclusion methods and their impact.
Task:
• Apply an inline style to make one heading red.
• Use an embedded <style> tag in the <head> to define body background.
• Link an external stylesheet styles.css and move all reusable styles there.
• Add comments in your CSS to label each section (/* Header styles */)

Code:
E1.html
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 1</title>

  <!-- Internal CSS -->
  <style>
    body{
      background-color:#f0f8ff;
    }
  </style>

  <!-- External CSS -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1 style="color:red;">Local Community Event Portal</h1>

  <div class="eventCard">
    Welcome to Community Events
  </div>

</body>
</html>
```
E1.css
```css
/* Event Card Styles */

.eventCard{
  border:1px solid black;
  padding:15px;
  background-color:white;
  width:250px;
```

}

Output:



Explanation:

- CSS can be applied using inline, internal, and external methods.
- External CSS improves reusability and maintenance.

Exercise 2

Problem Statement:
CSS Syntax and Comments
Scenario: You’ve joined a team and need to understand and maintain a large stylesheet.
Objective: Write clean, readable CSS with proper structure and comments.
Task:
• Create a section in styles.css with formatted rules and consistent indentation.
• Add descriptive comments above selectors.
• Example:

```
/* Style for main CTA button */
.cta-button {
background-color: #007BFF;
color: white;
}
```

Code:
E2.html

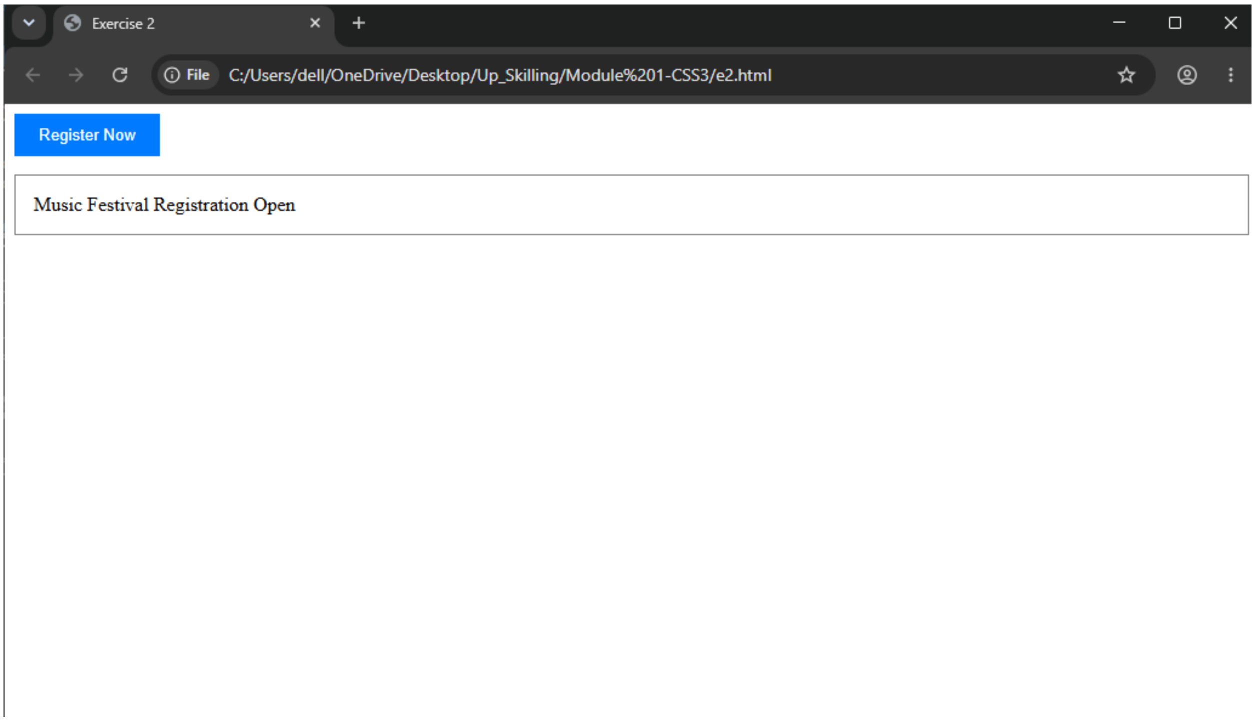
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 2</title>

  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <button class="cta-button">
    Register Now
  </button>

  <div class="eventCard">
    Music Festival Registration Open
  </div>

</body>
</html>
```

E2.css

```
/* Style for Main CTA Button */

.cta-button{
  background-color:#007BFF;
  color:white;
  padding:10px 20px;
  border:none;
  cursor:pointer;
}

/* Event Card Styling */

.eventCard{
```

```
    border:1px solid gray;
    padding:15px;
    margin-top:15px;
}
```

Output:



Explanation:

- CSS comments help developers understand stylesheet sections.
- Proper formatting makes CSS easier to maintain.

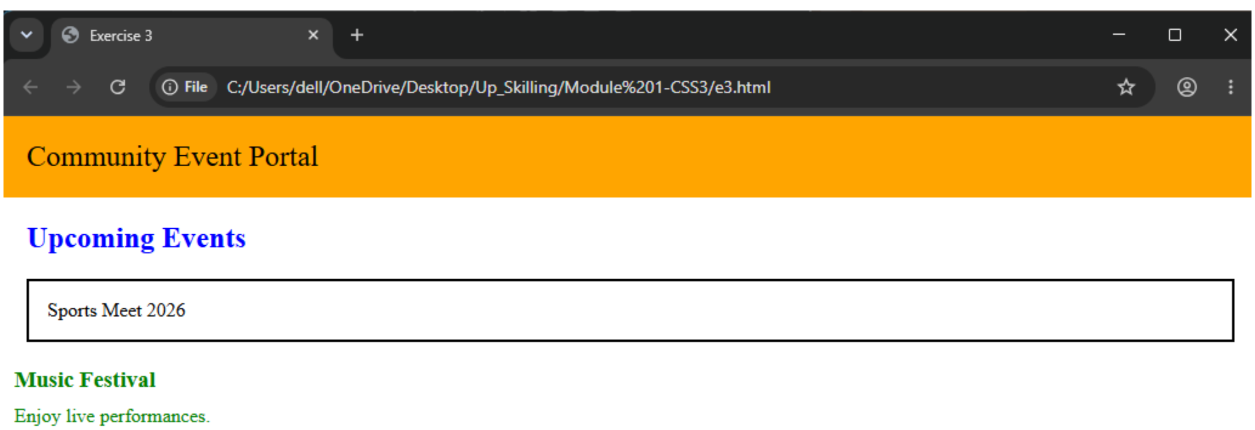
Exercise 3

Problem Statement:
Selectors Playground
Scenario: You need to style various elements based on IDs, classes, and element types.
Objective: Master different selector types.
Task:
• Use:
o Universal selector * to reset margin/padding
o Element selector to style all <h2>
o ID selector #mainHeader for the banner
o Class selector .eventCard for event containers
o Grouping selector for h3, p to style together

Code:
E3.html
```html
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Exercise 3</title>

   <link rel="stylesheet" href="styles.css">
</head>
<body>

   <div id="mainHeader">
      Community Event Portal
   </div>

   <h2>Upcoming Events</h2>

   <div class="eventCard">
      Sports Meet 2026
   </div>

   <h3>Music Festival</h3>

   <p>Enjoy live performances.</p>

</body>
</html>
```
E3.css
```css
/* Universal Selector */

*{
   margin:0;
   padding:0;
   box-sizing:border-box;
}

/* Element Selector */
```

```
h2{
   color:blue;
   margin:20px;
}

/* ID Selector */

#mainHeader{
   background-color:orange;
   padding:20px;
   font-size:24px;
}

/* Class Selector */

.eventCard{
   border:2px solid black;
   margin:20px;
   padding:15px;
}

/* Grouping Selector */

h3,p{
   color:green;
   margin:10px;
}
```

Output:



Explanation:

- Different selectors target elements in different ways.
- Grouping selectors apply common styles to multiple elements.

Exercise 4

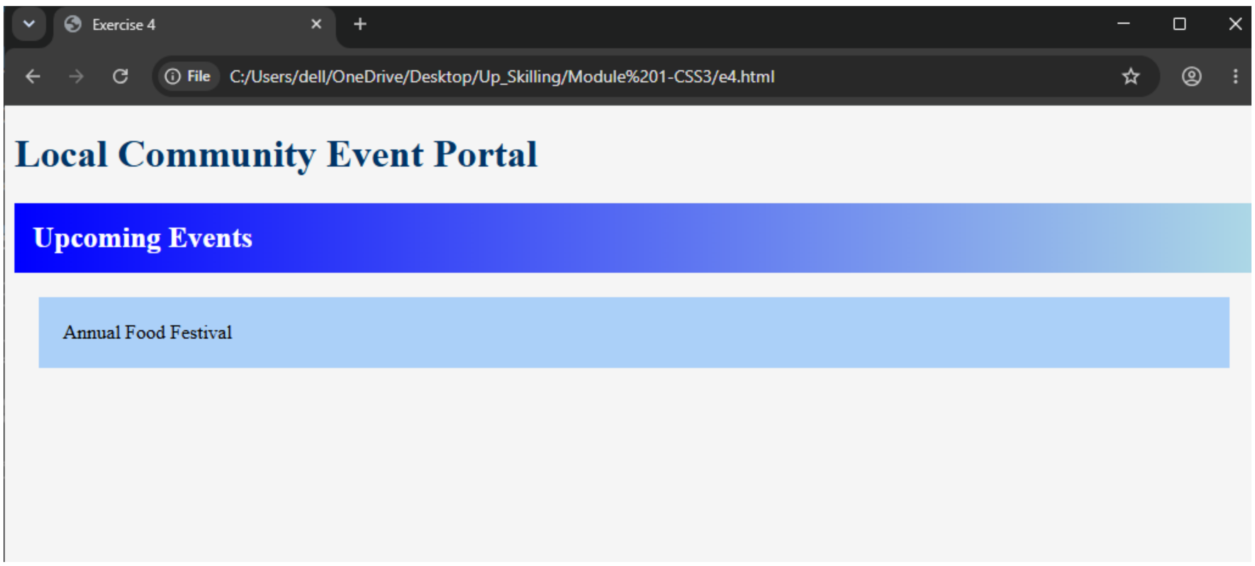
Problem Statement:
Color & Background Styling
Scenario: You're theming the portal based on a city council's branding.
Objective: Apply consistent colors and background visuals.
Task:
- Use HEX and RGBA for setting text and background colors
- Add a background image to the body with fallback color
- Apply gradients to section headers using background: linear-gradient(...)

Code:
E4.html
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 4</title>

  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1>Local Community Event Portal</h1>

  <h2 class="sectionHeader">
    Upcoming Events
  </h2>

  <div class="eventCard">
    Annual Food Festival
  </div>

</body>
</html>
```

E4.css
```css
body{
  background-color:#f5f5f5;
  background-image:url("background.jpg");
  background-size:cover;
}

h1{
  color:#003366;
}

.eventCard{
  background-color:rgba(0,123,255,0.3);
  padding:20px;
  margin:20px;
}
```

```
.sectionHeader{
   background:linear-gradient(to right, blue, lightblue);
   color:white;
   padding:15px;
}
```

Output:



Explanation:

- HEX and RGBA colors provide flexible color styling.
- Gradients and background images improve visual appearance.

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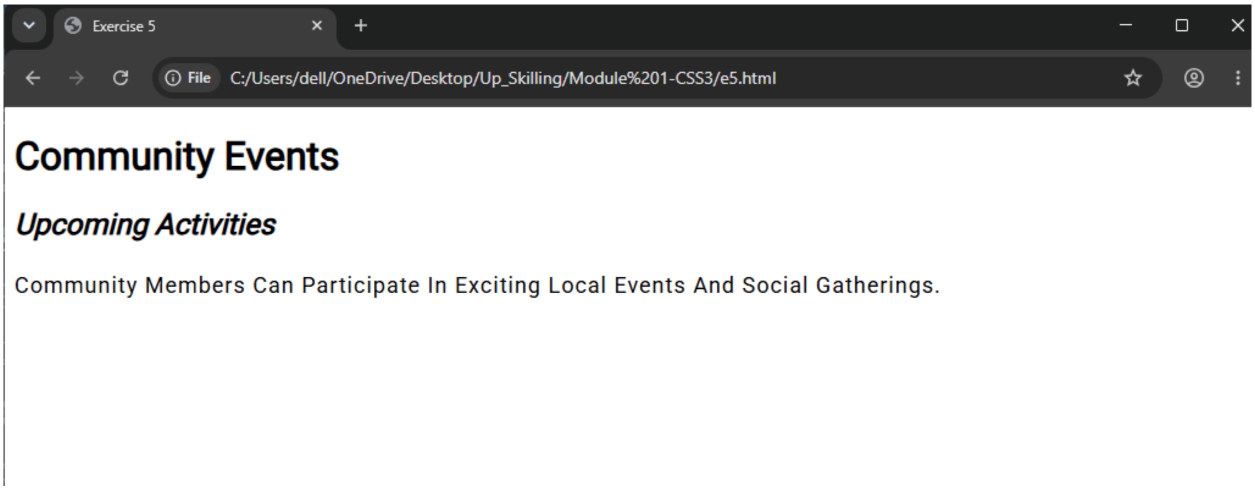
Exercise 5

Problem Statement:
Typography: Fonts and Text
Scenario: The marketing team wants more appealing fonts and better readability.
Objective: Enhance textual appearance using CSS properties.
Task:
• Use @import or <link> to include a Google Font
• Set font-family, font-size, font-style, font-weight in different sections
• Use text-align, text-transform, letter-spacing, line-height on descriptions
Code:
E5.html
```html
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Exercise 5</title>

   <link href="https://fonts.googleapis.com/css2?family=Roboto&display=swap" rel="stylesheet">

   <link rel="stylesheet" href="styles.css">
</head>
<body>

   <h1>Community Events</h1>

   <h2>Upcoming Activities</h2>

   <p class="description">
      Community members can participate in exciting local events and social gatherings.
   </p>

</body>
</html>
```
E5.css
```css
body{
   font-family:'Roboto',sans-serif;
}

h1{
   font-size:32px;
   font-weight:bold;
}

h2{
   font-style:italic;
}

.description{
   text-align:justify;
   text-transform:capitalize;
```

```
  letter-spacing:1px;
  line-height:1.8;
  font-size:18px;
}
```

Output:



Explanation:

- Typography properties improve readability and presentation.
- Google Fonts provide modern and attractive font styles.